

Temporal dependence - headache

Patients 61; gap transitions 208 (max 37 days); missing-day fraction 0.0955.

Markov: $P(\text{attack tomorrow} \mid \text{attack today}) = 0.397$ vs 0.179 without (risk ratio 2.2). Self-excitation lag-1 OR 2.44 (trigger-controlled).

Pre-registered primary tests (raw p)

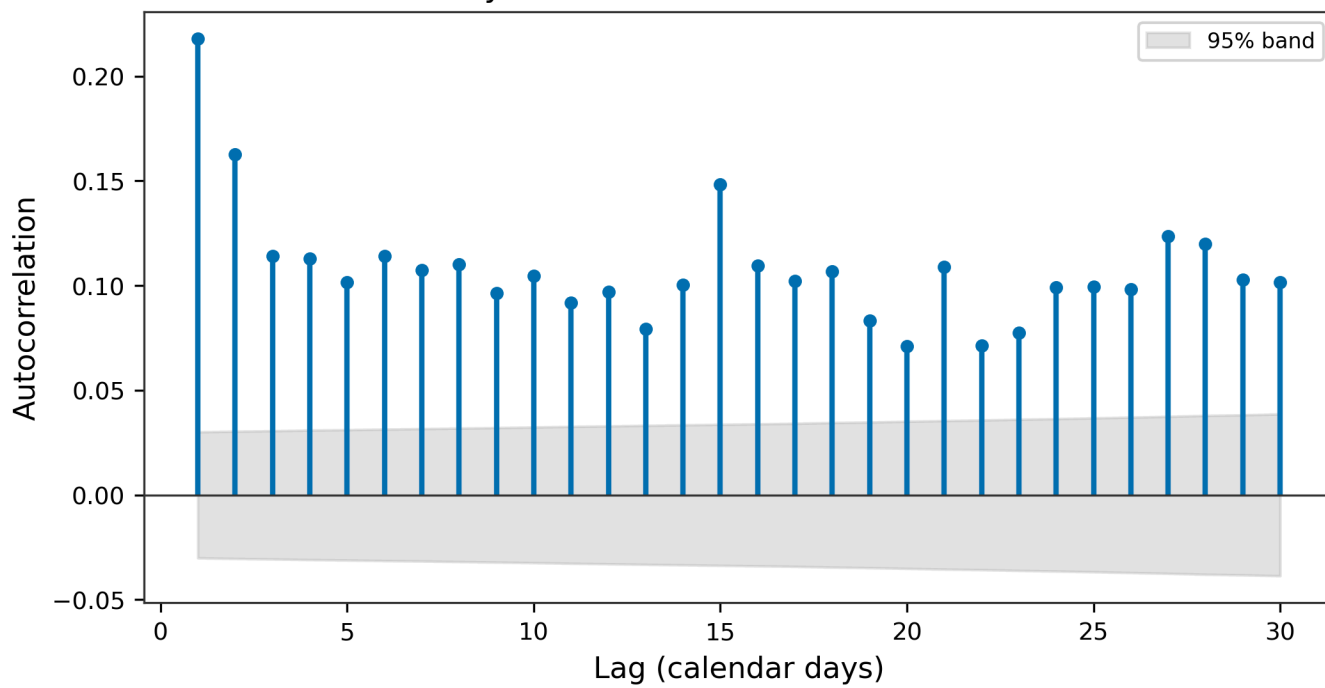
test	p
lag-1 Markov transition	1.81e-45
self-excitation LR (vs triggers-only)	1.10e-43
day-of-week omnibus	5.21e-01

Exploratory ACF lags (BH-FDR adjusted)

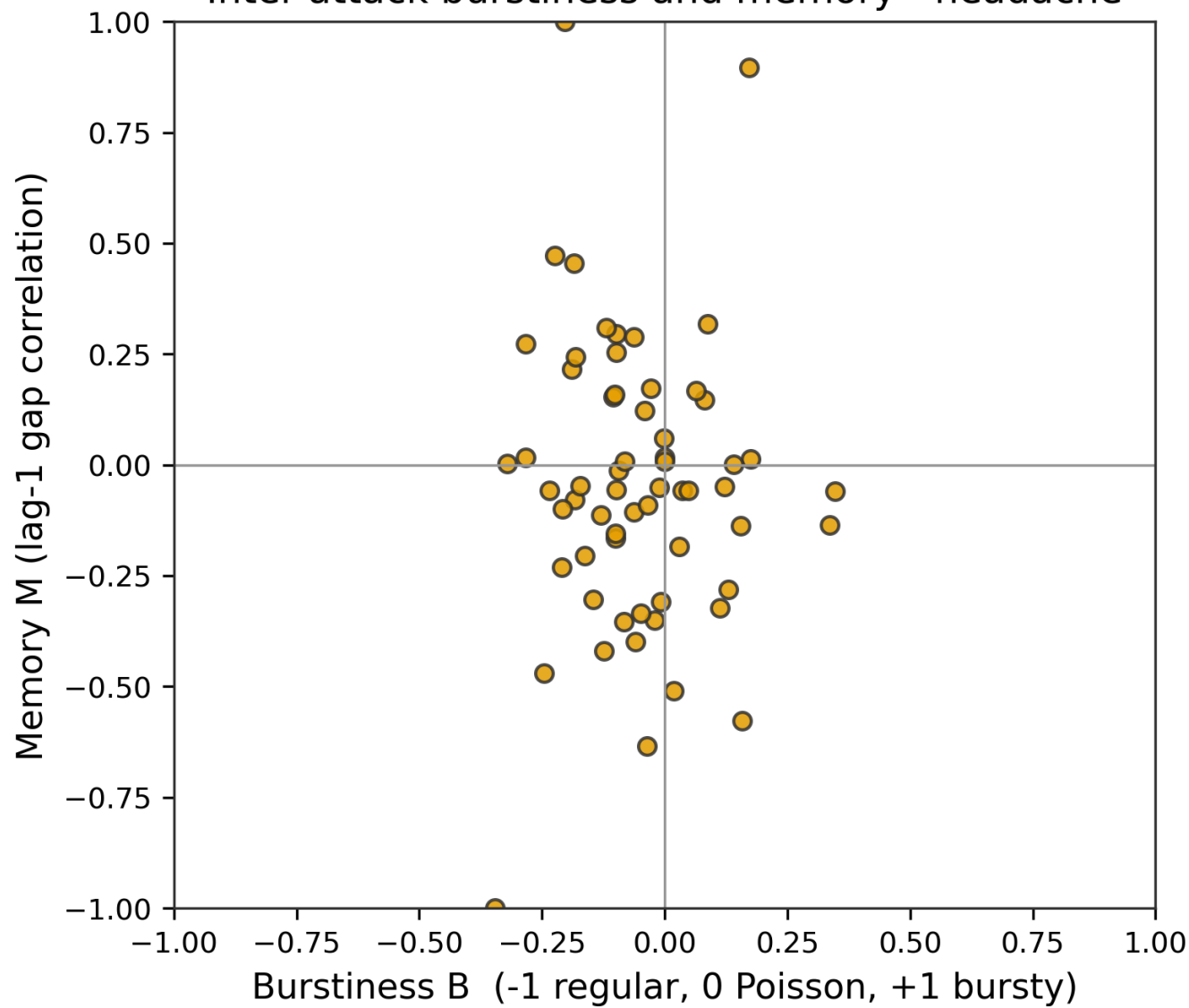
lag	p	p_adj
ACF lag 2	9.38e-26	1.22e-24
ACF lag 3	2.63e-13	1.71e-12
ACF lag 4	8.77e-13	3.44e-12
ACF lag 5	1.57e-10	2.92e-10
ACF lag 6	1.06e-12	3.44e-12
ACF lag 7	2.56e-11	5.56e-11
ACF lag 8	1.24e-11	3.22e-11
ACF lag 9	3.83e-09	5.08e-09
ACF lag 10	2.49e-10	4.05e-10
ACF lag 11	3.44e-08	3.72e-08
ACF lag 12	7.18e-09	8.49e-09
ACF lag 13	2.82e-06	2.82e-06
ACF lag 14	3.91e-09	5.08e-09

Figures

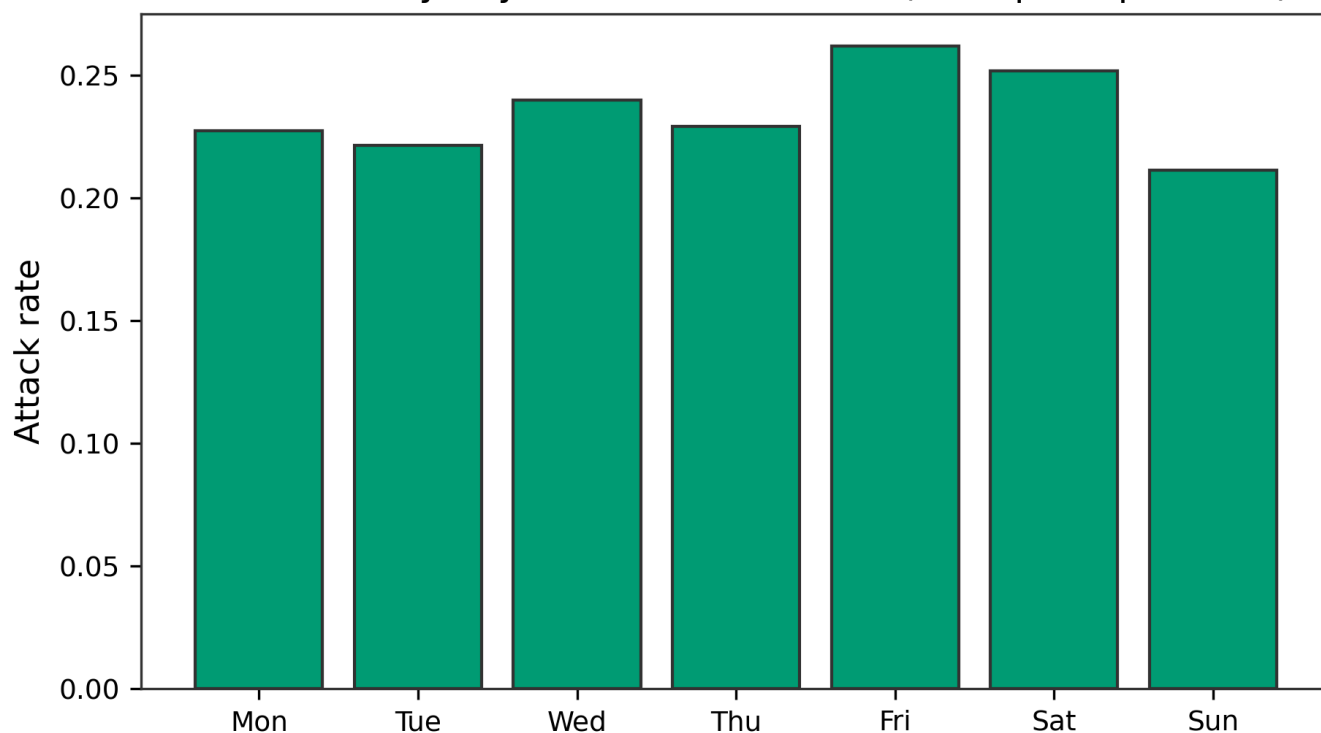
Daily attack autocorrelation - headache



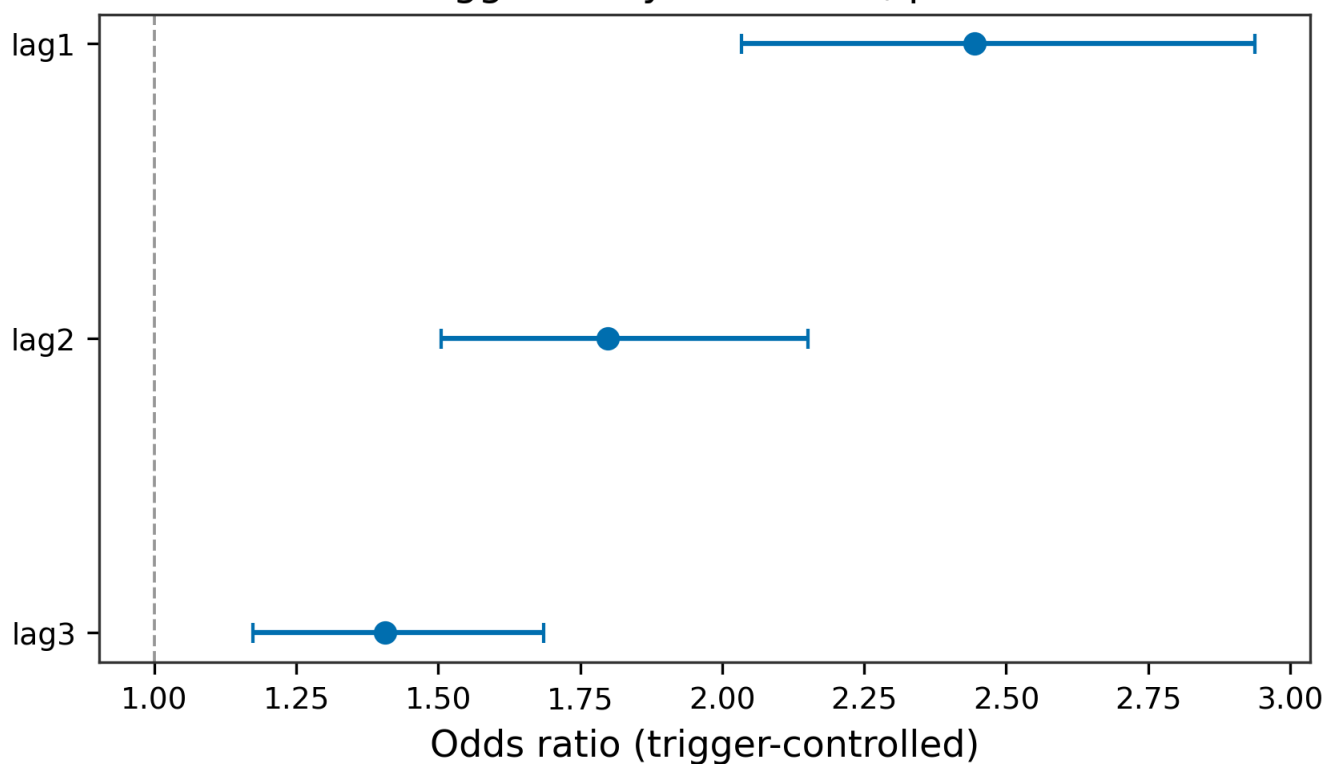
Inter-attack burstiness and memory - headache



Attack rate by day of week - headache (chi-square $p = 0.52$)



Attack-lag self-excitation - headache
LR vs triggers-only: $\chi^2=203$, $p=1.1e-43$



First-order transition - headache (RR=2.2, p=1.8e-45)

